

HE2003 Econometrics I

Quiz Practice 2 (Solutions)

AY 2026/2027, Semester 1

Quantitative Research Society @NTU

October 1, 2026

Overview

- **Coverage:** Lectures 1–5 and Lecture 6 slides 1–3 (large-sample distribution of $\hat{\beta}_1$ and its standard error); no t-tests, confidence intervals, p-values or Stata. This paper is slightly harder than Practice 1: it includes an estimator-comparison-with-weights derivation, a Least Squares Assumption 1 counterexample, a $TSS/SSR/R^2$ computation, and a sampling-distribution question about the difference of two independent sample means.
- **Format:** 5 questions, 50 marks total, approximately 60 minutes.
- **Question mix:** Q1 derivation (efficient combination of two independent unbiased estimators, with weights); Q2 regression-output interpretation (prediction, change in prediction, and $TSS/SSR/R^2$); Q3 sampling distribution of a difference of two independent sample means (mean, variance, a standardised probability, and the effect of quadrupling n on the standard deviation); Q4 binary regressor (group means, and a common-slope/different-intercepts extension with a second regressor); Q5 conceptual (selection bias, a Least Squares Assumption 1 counterexample, Type I/II trade-off).
- **Standard normal table** ($\Phi(z) = P(Z \leq z)$ for $Z \sim N(0, 1)$), needed in Question 3:

z	0.50	1.00	1.28	1.50	1.645	1.96	2.00	2.50	2.58	3.00
$\Phi(z)$	0.6915	0.8413	0.8997	0.9332	0.9500	0.9750	0.9772	0.9938	0.9951	0.9987

Use $P(Z > z) = 1 - \Phi(z)$ and, by symmetry, $\Phi(-z) = 1 - \Phi(z)$.

Instructions

Answer **all** questions. Show all working clearly; answers without justification may receive zero marks.

Contents

Question 1	(10 Marks)	3
Solution		3
Method 1: Calculus (first-order condition)		3
Method 2: Completing the square		3
Marking scheme (10 marks)		4
Question 2	(10 Marks)	5
Solution		5
Method 1: Direct approach		5
Marking scheme (10 marks)		5
Question 3	(10 Marks)	7
Solution		7
Method 1: Direct approach		7
Marking scheme (10 marks)		7
Question 4	(10 Marks)	9
Solution		9
Method 1: Direct approach		9
Marking scheme (10 marks)		10
Question 5	(10 Marks)	11
Solution		11
Method 1: Direct approach		11
Marking scheme (10 marks)		12

Question 1

(10 Marks)

A government agency wants to estimate the true population mean μ of weekly working hours in a sector. It combines two **independent** unbiased surveys: survey 1 has $\hat{\mu}_1$ with $\text{Var}(\hat{\mu}_1) = v_1$, and survey 2 has $\hat{\mu}_2$ with $\text{Var}(\hat{\mu}_2) = v_2$, where both are unbiased for μ , i.e. $E[\hat{\mu}_1] = E[\hat{\mu}_2] = \mu$. For a weight $w \in [0, 1]$, consider the combined estimator

$$\hat{\mu}_w = w\hat{\mu}_1 + (1 - w)\hat{\mu}_2.$$

- (a) (3 marks) Show that $\hat{\mu}_w$ is unbiased for μ for *every* choice of $w \in [0, 1]$.
- (b) (4 marks) Using independence of $\hat{\mu}_1$ and $\hat{\mu}_2$, derive $\text{Var}(\hat{\mu}_w)$ as a function of w , v_1 , and v_2 .
- (c) (3 marks) Find the weight w^* that minimizes $\text{Var}(\hat{\mu}_w)$, and show that it can be written as $w^* = \frac{v_2}{v_1 + v_2}$ (i.e. the more precise survey, with the smaller variance, receives the larger weight).

Solution

Method 1: Calculus (first-order condition)

(a)

$$\begin{aligned} E[\hat{\mu}_w] &= wE[\hat{\mu}_1] + (1 - w)E[\hat{\mu}_2] = w\mu + (1 - w)\mu && \text{(linearity)} \\ \implies &\boxed{E[\hat{\mu}_w] = \mu \text{ for all } w \in [0, 1].} \end{aligned}$$

(b)

$$\begin{aligned} \text{Var}(\hat{\mu}_w) &= w^2v_1 + (1 - w)^2v_2 + 2w(1 - w)\text{Cov}(\hat{\mu}_1, \hat{\mu}_2) && (\text{Var}(cX) = c^2\text{Var}(X)) \\ \implies &\boxed{\text{Var}(\hat{\mu}_w) = w^2v_1 + (1 - w)^2v_2.} && (\text{Cov} = 0 \text{ by independence}) \end{aligned}$$

(c)

$$\begin{aligned} \frac{d}{dw} \text{Var}(\hat{\mu}_w) &= 2wv_1 - 2(1 - w)v_2 = 0 && \text{(FOC)} \\ \frac{d^2}{dw^2} \text{Var}(\hat{\mu}_w) &= 2(v_1 + v_2) > 0 && \text{(SOC: minimum)} \\ \implies &w(v_1 + v_2) = v_2 \\ \implies &\boxed{w^* = \frac{v_2}{v_1 + v_2}.} && \text{(smaller } v, \text{ larger weight)} \end{aligned}$$

Method 2: Completing the square

$$\begin{aligned} \text{Var}(\hat{\mu}_w) &= (v_1 + v_2) \left(w - \frac{v_2}{v_1 + v_2} \right)^2 + \frac{v_1v_2}{v_1 + v_2} \\ &\geq \frac{v_1v_2}{v_1 + v_2} \text{ (since } v_1 + v_2 > 0), \text{ equality iff } w = \frac{v_2}{v_1 + v_2} \implies \boxed{w^* = \frac{v_2}{v_1 + v_2}} \end{aligned}$$

Marking scheme (10 marks)**(a) (3 marks) Unbiasedness**

- 1 mark: correctly applies linearity of expectation to $\hat{\mu}_w$.
- 1 mark: correctly substitutes $E[\hat{\mu}_1] = E[\hat{\mu}_2] = \mu$.
- 1 mark: correct conclusion $E[\hat{\mu}_w] = \mu$ for every w .

(b) (4 marks) Variance

- 1 mark: correctly expands $\text{Var}(w\hat{\mu}_1 + (1 - w)\hat{\mu}_2)$ including the covariance term.
- 1 mark: correctly sets $\text{Cov}(\hat{\mu}_1, \hat{\mu}_2) = 0$ by independence.
- 1 mark: correct use of $\text{Var}(cX) = c^2 \text{Var}(X)$ for both terms.
- 1 mark: correct final expression $\text{Var}(\hat{\mu}_w) = w^2 v_1 + (1 - w)^2 v_2$.

(c) (3 marks) Optimal weight

- 1 mark: correct first-order condition (or completed-square form).
- 1 mark: correct algebra solving for w^* .
- 1 mark: correct final answer $w^* = v_2/(v_1 + v_2)$ with a valid remark on precision-weighting (or second-order/positivity check).

Question 2**(10 Marks)**

PUB (Singapore's national water agency) commissions a study of $n = 200$ households and regresses monthly water consumption (in cubic metres, m^3) on household size (number of residents). The estimated regression is

$$\widehat{\text{Water}}_i = 3.8 + 5.4 \text{Size}_i, \quad \text{SE}(\hat{\beta}_1) = 1.2.$$

- (3 marks) What is the regression's predicted monthly water consumption for a household of 5 residents?
- (2 marks) A household grows from 3 to 6 residents. What is the regression's predicted change in monthly water consumption?
- (3 marks) For this regression, $\text{TSS} = 5000$ and $\text{SSR} = 1500$ (in units of $(\text{m}^3)^2$). Compute ESS and R^2 , and interpret R^2 in words.
- (2 marks) Interpret the slope coefficient with units, and comment on whether the intercept is economically meaningful here.

Solution**Method 1: Direct approach**

(a)

$$\widehat{\text{Water}} = 3.8 + 5.4(5) \implies \boxed{\widehat{\text{Water}} = 30.8 \text{ m}^3/\text{month}.}$$

(b)

$$\Delta \widehat{\text{Water}} = \hat{\beta}_1 \Delta \text{Size} = 5.4(6 - 3) \implies \boxed{\Delta \widehat{\text{Water}} = 16.2 \text{ m}^3/\text{month (an increase).}}$$

(c)

$$\begin{aligned} \text{ESS} &= \text{TSS} - \text{SSR} = 5000 - 1500 \quad (\text{TSS} = \text{ESS} + \text{SSR}) \\ R^2 &= \text{ESS}/\text{TSS} = 3500/5000 = 0.7 \quad (70\% \text{ of the variation explained by Size}) \\ \implies &\boxed{\text{ESS} = 3500 (\text{m}^3)^2, \quad R^2 = 0.70.} \end{aligned}$$

(d) $\hat{\beta}_0 = 3.8$ is the prediction at $\text{Size} = 0$; no household has 0 residents.

$$\boxed{\text{Slope: } +5.4 \text{ m}^3/\text{month per resident; intercept not economically meaningful.}}$$

Marking scheme (10 marks)

(a) **(3 marks)** Predicted consumption

- 1 mark: correct substitution.
- 1 mark: correct arithmetic.
- 1 mark: correct units.

(b) **(2 marks)** Predicted change

- 1 mark: correct method/arithmetic (5.4×3).
- 1 mark: correctly stated with units and direction.

(c) **(3 marks)** ESS and R^2

- 1 mark: correct use of $TSS = ESS + SSR$ to get $ESS = 3500$.
- 1 mark: correct $R^2 = ESS/TSS = 0.70$.
- 1 mark: correct interpretation of R^2 as the share of variation in water consumption explained by household size.

(d) **(2 marks)** Interpretation

- 1 mark: correct slope interpretation with units.
- 1 mark: correct comment on the intercept's lack of economic meaning.

Question 3**(10 Marks)**

Two independent surveys estimate average weekly online-shopping spending (in SGD) in two regions of a large Southeast Asian city. Region A: sample size $n_A = 100$, sample mean $\bar{Y}_A$, with $\text{Var}(\bar{Y}_A) = 16$. Region B: sample size $n_B = 150$, sample mean $\bar{Y}_B$, with $\text{Var}(\bar{Y}_B) = 9$. The two samples are drawn independently, and the true population means satisfy $\mu_A - \mu_B = 10$.

- (a) (3 marks) Derive $E[\bar{Y}_A - \bar{Y}_B]$ and $\text{Var}(\bar{Y}_A - \bar{Y}_B)$, using the independence of the two samples.
- (b) (4 marks) In large samples, $\bar{Y}_A - \bar{Y}_B$ is approximately normally distributed. Standardise and use the standard normal table in the Overview to compute $P(\bar{Y}_A - \bar{Y}_B > 17.5)$.
- (c) (3 marks) Suppose both sample sizes are quadrupled ($n'_A = 400$, $n'_B = 600$) while the population variances of individual spending stay the same. Explain what happens to $\text{Var}(\bar{Y}_A)$ and $\text{Var}(\bar{Y}_B)$, and hence find the new standard deviation of $\bar{Y}_A - \bar{Y}_B$.

Solution**Method 1: Direct approach**

(a)

$$\begin{aligned}
 E[\bar{Y}_A - \bar{Y}_B] &= \mu_A - \mu_B = 10 && \text{(linearity)} \\
 \text{Var}(\bar{Y}_A - \bar{Y}_B) &= \text{Var}(\bar{Y}_A) + \text{Var}(\bar{Y}_B) - 2 \text{Cov}(\bar{Y}_A, \bar{Y}_B) \\
 &= 16 + 9 - 0 = 25 && \text{(Cov = 0 by independence)} \\
 \implies &\boxed{E[\bar{Y}_A - \bar{Y}_B] = 10, \quad \text{Var}(\bar{Y}_A - \bar{Y}_B) = 25.}
 \end{aligned}$$

(b)

$$\begin{aligned}
 P(\bar{Y}_A - \bar{Y}_B > 17.5) &= P\left(\frac{(\bar{Y}_A - \bar{Y}_B) - 10}{5} > \frac{17.5 - 10}{5}\right) && \text{(s.d. = 5)} \\
 &= P(Z > 1.5) = 1 - \Phi(1.5) = 1 - 0.9332 && \text{(table)} \\
 \implies &\boxed{P(\bar{Y}_A - \bar{Y}_B > 17.5) = 0.0668.}
 \end{aligned}$$

(c)

$$\begin{aligned}
 \text{Var}(\bar{Y}'_A) &= \frac{16}{4} = 4, \quad \text{Var}(\bar{Y}'_B) = \frac{9}{4} = 2.25 && (\text{Var}(\bar{Y}) = \sigma^2/n, n \rightarrow 4n) \\
 \text{Var}(\bar{Y}'_A - \bar{Y}'_B) &= 4 + 2.25 = 6.25 \implies \text{s.d.} = \sqrt{6.25} = 2.5 && \text{(independence)}
 \end{aligned}$$

$$\boxed{\text{New Var}(\bar{Y}'_A - \bar{Y}'_B) = 6.25, \quad \text{new s.d.} = 2.5 \text{ (half the original s.d. of 5).}}$$

Marking scheme (10 marks)(a) **(3 marks)** Mean and variance

- 1 mark: correct $E[\bar{Y}_A - \bar{Y}_B] = 10$ by linearity of expectation.
- 1 mark: correctly sets $\text{Cov}(\bar{Y}_A, \bar{Y}_B) = 0$ by independence.
- 1 mark: correct final variance 25.

(b) **(4 marks)** Probability

- 1 mark: correct standard deviation $\sqrt{25} = 5$.
- 1 mark: correct standardisation to $z = 1.5$.
- 1 mark: correctly reads $\Phi(1.5) = 0.9332$ from the table.
- 1 mark: correct final probability 0.0668.

(c) **(3 marks)** Effect of quadrupling n

- 1 mark: correctly identifies $\text{Var}(\bar{Y}_A)$, $\text{Var}(\bar{Y}_B)$ each fall by a factor of 4.
- 1 mark: correct new variance of the difference, 6.25.
- 1 mark: correct new standard deviation 2.5, with correct link to it being half the original.

Question 4

(10 Marks)

A study of $n = 160$ hawker stalls in Singapore regresses average daily revenue (in SGD) on D_i , a dummy variable equal to 1 if the stall accepts e-payment and 0 if it is cash-only ($n_0 = 60$ cash-only, $n_1 = 100$ e-payment). The estimated regression is

$$\widehat{\text{Revenue}}_i = 520 + 85 D_i, \quad \text{SE}(\hat{\beta}_1) = 28.$$

- (a) (3 marks) State the average daily revenue predicted by the regression for cash-only stalls and for e-payment stalls.
- (b) (2 marks) Give an economic interpretation of $\hat{\beta}_1 = 85$.
- (c) (3 marks) The platform's data team re-estimates the model to control for the number of hours each stall operates per day, Hours_i , assuming a *common slope* on Hours_i for both groups (i.e. the effect of an extra hour of operation is the same for cash-only and e-payment stalls):

$$\widehat{\text{Revenue}}_i = 150 + 85 D_i + 45 \text{Hours}_i.$$

Write the separate fitted lines for cash-only stalls ($D = 0$) and e-payment stalls ($D = 1$) as functions of Hours_i . State which group is the *base group*, and give an economic interpretation of the coefficient 85 on D_i in this extended model.

- (d) (2 marks) Explain the *dummy variable trap*: why can the regression in part (c) *not* also include a second dummy $C_i = 1 - D_i$ (an indicator for cash-only) alongside D_i and the intercept? State the general rule for how many dummy variables to include for a categorical variable with g mutually exclusive groups.

Solution

Method 1: Direct approach

(a)

$$D = 0 : 520 + 85(0) = 520$$

$$D = 1 : 520 + 85(1) = 605$$

$$\implies \boxed{\text{Cash-only: SGD 520/day; E-payment: SGD 605/day.}}$$

(b) D binary $\implies \hat{\beta}_1 =$ difference in mean daily revenue between groups.

$$\boxed{\text{E-payment stalls earn, on average, SGD 85/day more than cash-only stalls.}}$$

(c) Base group: cash-only ($D_i = 0$). Common slope 45; holding Hours fixed, e-payment stalls earn SGD 85/day more (vertical gap between the parallel lines).

$$D = 0 : 150 + 45 \text{Hours}$$

$$D = 1 : (150 + 85) + 45 \text{Hours}$$

$$\implies \boxed{\begin{array}{l} \text{Cash-only: } \widehat{\text{Revenue}} = 150 + 45 \text{Hours} \\ \text{E-payment: } \widehat{\text{Revenue}} = 235 + 45 \text{Hours} \end{array}}$$

(d) • $D_i + C_i = 1 =$ intercept column $\implies$ perfect multicollinearity, no unique OLS solution.

- Rule: g groups $\implies g - 1$ dummies plus the intercept; the omitted group is the base.

$D_i + C_i = 1$ for all $i \implies$ exact collinearity with the intercept;
use only $g - 1$ dummies for g groups.

Marking scheme (10 marks)

(a) **(3 marks)** Group means

- 1 mark: correct method.
- 1 mark: correct cash-only mean (520).
- 1 mark: correct e-payment mean (605).

(b) **(2 marks)** Interpretation

- 1 mark: recognizes $\hat{\beta}_1$ as the difference in group means.
- 1 mark: correct direction/magnitude with units.

(c) **(3 marks)** Common-slope, different-intercepts model

- 1 mark: correct fitted line for cash-only stalls.
- 1 mark: correct fitted line for e-payment stalls, and correctly identifies cash-only as the base group.
- 1 mark: correct interpretation of the coefficient 85 holding Hours fixed (parallel-lines / vertical-shift interpretation).

(d) **(2 marks)** Dummy variable trap

- 1 mark: correctly explains the exact collinearity $D_i + C_i = 1$ with the intercept and why it breaks OLS.
- 1 mark: correctly states the general rule ($g - 1$ dummies for g groups, one omitted as base).

Question 5

(10 Marks)

- (a) (4 marks) A ride-hailing platform in a large Southeast Asian city regresses a driver's weekly earnings on the number of hours the driver chose to be online (available for trips), using data from drivers who freely set their own hours. The platform's data team finds a strong positive coefficient and concludes that staying online longer *causes* higher weekly earnings on a per-hour basis. Explain, using Least Squares Assumption 1, why this conclusion may not be valid, giving a specific unobserved factor and the likely direction of the resulting bias.
- (b) (3 marks) **Least Squares Assumption 1 counterexample.** Let $X \in \{0, 1\}$ be a binary regressor with $P(X = 0) = P(X = 1) = 0.5$, and let u be an error term with $E[u | X = 0] = -2$ and $E[u | X = 1] = 2$.
- Show that the *unconditional* mean $E[u] = 0$.
 - Explain why Least Squares Assumption 1 is nonetheless violated, and briefly describe (in words or with a simple sketch) what this would look like: two groups whose regression errors are centred at different, non-zero values even though the overall average error is zero.
- (c) (3 marks) A public-health agency screens travellers for a disease outbreak using a rapid test, classifying each traveller as “positive” (quarantine) or “negative” (release) based on a biomarker threshold. Define, in this context, a Type I error and a Type II error, and explain the trade-off the agency faces if it lowers the biomarker threshold required to be classified “positive”.

Solution

Method 1: Direct approach

- (a)
- Least Squares Assumption 1 needs $E[u_i | \text{Hours}_i] = 0$; drivers choose hours, so unobserved skill (route efficiency) in u_i is correlated with Hours.
 - Skilled drivers stay online longer and earn more per hour $\implies \text{Cov}(\text{Hours}_i, u_i) > 0$.

$$E[u | \text{Hours}] \neq 0 \text{ (self-selection on driver skill)} \implies \hat{\beta}_1 \text{ upward biased.}$$

- (b) (i)

$$\begin{aligned} E[u] &= P(X = 0)E[u | X = 0] + P(X = 1)E[u | X = 1] \quad (\text{total expectation}) \\ &= 0.5(-2) + 0.5(2) \implies E[u] = 0. \end{aligned}$$

- (ii) $E[u | X = 0] = -2 \neq 0$, $E[u | X = 1] = 2 \neq 0$: Least Squares Assumption 1 fails (errors centred at ∓ 2 in the two groups).

$$E[u] = 0 \text{ does not imply } E[u | X] = 0 \text{ for all } X: \\ \text{Assumption 1 needs the conditional statement.}$$

- (c)
- Type I: a healthy traveller is classified “positive” (unnecessary quarantine).
 - Type II: an infected traveller is classified “negative” (released, spreads disease).

- Lower threshold: more positives, so Type II falls and Type I rises (one threshold drives both).

Lower threshold: $\downarrow$ Type II error (fewer missed cases),
 $\uparrow$ Type I error (more healthy travellers quarantined).

Marking scheme (10 marks)

(a) **(4 marks)** Selection bias

- 1 mark: correctly states Assumption 1 as the relevant condition.
- 1 mark: identifies a specific, plausible confounding factor (e.g. driver skill/local knowledge).
- 1 mark: explains why the confounder correlates with hours chosen (self-selection).
- 1 mark: correct direction of bias (upward) with valid justification.

(b) **(3 marks)** Assumption 1 counterexample

- 1 mark: correct computation $E[u] = 0$ via the law of total expectation.
- 1 mark: correctly identifies that $E[u | X = 0] \neq 0$ and $E[u | X = 1] \neq 0$ violate Assumption 1.
- 1 mark: valid explanation/description that unconditional $E[u] = 0$ does not imply conditional mean independence.

(c) **(3 marks)** Type I/II trade-off

- 1 mark: correct Type I error definition in context.
- 1 mark: correct Type II error definition in context.
- 1 mark: correct description of the trade-off when the threshold is lowered.